Vansh Gupta

+91 9289011325 | vanshgupta7266@gmail.com | LinkedIn | GitHub
CodeChef 4-Star (max 1856) | Codeforces Pupil (max 1340) | Bhopal, India

Professional Summary

Third-year B.Tech student at IIIT Bhopal (CPI 8.49/10) building production ML systems at the intersection of generative models, multimodal AI, and applied research. Currently ML Intern at Moleculyst developing multimodal Video LLM and Vision-Language Model inference pipelines with structured evaluation loops. Shipped end-to-end ML systems spanning a 5-agent LangGraph + Gemini bias-mitigation pipeline (Equitas AI), an LSTM + XGBoost pharmaceutical batch-quality forecaster with SHAP attribution (PharmAI, >90% accuracy), and a reinforcement learning environment for LLaMA 3.1 (Triage AI). Strong probability, linear algebra, and statistical foundations from JEE-level math; actively deepening into mathematical optimization and generative modeling theory. Excited to apply generative ML and agentic AI to ambitious drug-discovery research alongside a feedback-driven engineering team.

Education

IIIT Bhopal Bhopal, Madhya Pradesh, India
B.Tech in Information Technology (CS & Engineering Curriculum) | CPI: 8.49 / 10 Aug 2024 – May 2028

- **Relevant Coursework**: Data Structures & Algorithms, Object-Oriented Programming, Linear Algebra, Probability & Statistics, Discrete Mathematics, Operating Systems, Database Management Systems, Computer Networks

Technical Skills

ML / Deep Learning: PyTorch, TensorFlow, Scikit-learn, Transformers, LSTM, XGBoost, SHAP, Generative Models, Reinforcement Learning, RAG

LLM / Multimodal Stack: LangChain, LangGraph, VLMs, Multimodal LLMs, Prompt Engineering, Gemini API, LLaMA 3.1

Research & Math Foundations: Probability Theory, Linear Algebra, Statistical Modeling, Mathematical Optimization, Time/Space Complexity Analysis

Languages: Python, C++, C, SQL

Engineering & Infrastructure: FastAPI, Docker, Git/GitHub, REST API Design, Software Testing, Pydantic, OpenEnv

Cloud & Deployment: AWS, GCP (Cloud Run, Vertex AI), Scalable Pipeline Architecture, Observability, Monitoring

Competitive Programming (C++): CodeChef 4-Star (max 1856), Codeforces Pupil (max 1340) – DP, graphs, segment trees, combinatorics

Experience

ML Intern Feb 2026 – Present
Moleculyst Remote

- Designing and shipping **multimodal Video LLM inference pipelines** in Python – integrating vision-language architectures, preprocessing stages, and structured evaluation loops – delivering Moleculyst's core inference system for structured understanding across complex operational video datasets.
- Built reproducible **evaluation frameworks** that standardized quality assessment across model versions, enabling consistent benchmarking and faster iteration on model and prompt changes.
- Expanding scope into **applied quantitative research** – probability theory, mathematical optimization, statistical modeling – to inform model selection, evaluation design, and uncertainty quantification across generative ML experiments.
- Collaborating with cross-functional stakeholders to iterate on technical designs, capture user requirements, and deliver software on schedule in a cooperative team environment.

Projects

PharmAI – Pharmaceutical Batch Quality Forecasting *Python, LSTM, XGBoost, SHAP, Scikit-learn* 2025

- Engineered an **LSTM + XGBoost ensemble** for pharmaceutical batch quality prediction trained on historical process data, achieving **>90% accuracy** on held-out validation batches – directly applicable to drug-manufacturing quality and process-development workflows.
- Implemented **SHAP-driven anomaly attribution** surfacing the top contributing process variables per flagged batch, enabling interpretable failure-mode analysis for chemists and process engineers.

Equitas AI – Multi-Agent Generative Bias Mitigation *Python, LangGraph, FastAPI, Gemini API, Vertex AI, GCP* 2025

- Designed a **5-agent autonomous LangGraph pipeline** for dataset bias auditing, orchestrating iterative mitigation loops that converged at Disparate Impact Ratio (DIR) ≥ 0.80 – the U.S. legal fairness threshold – across all evaluated datasets.
- Integrated **SHAP explainability matrices with Gemini-driven counterfactual narration**, surfacing row-level bias reasoning through an interactive review interface usable by non-technical stakeholders.

- Deployed a keyless FastAPI backend on Google Cloud Run via Vertex AI – fully serverless, autoscaling, no stored secrets.

Triage AI – Reinforcement Learning Environment for LLM Evaluation *Python, FastAPI, OpenEnv, Pydantic, Docker, LLaMA 3.1* 2025

- Architected a **reinforcement learning environment** for automated email classification, implementing structured OpenEnv state-machine cycles (reset / step / observe) with multi-level prompt-engineering chains, achieving **0.71 / 1.0 reward accuracy** on LLaMA 3.1 (8B).
- Containerized the full system with Docker and enforced API contracts with Pydantic – reproducible, environment-isolated deployment that can be reliably reset and re-evaluated across test cycles.

Moleculyst Studio – Multimodal Agentic Image-Editing Workspace *Python, Gemini API, VLMs, OpenCV, Docker* 2025

- Built a multi-step **agentic editing pipeline** composing VLM-based grounding, targeted edit operators, and an automated critique loop – replacing manual step-by-step review with a single-command end-to-end workflow.
- Streamed structured reasoning traces, tool calls, and per-step quality scores into a unified monitoring view, enabling engineers to inspect and intervene at any stage of the pipeline.

Achievements

- **Meta × PyTorch OpenEnv Hackathon – Grand Finalist** (Bangalore, 2025): Top-tier ranking out of 52,000+ global participants for shipping a production-ready reinforcement learning environment.
- **CodeBidz Hackathon – 1st Place**: Won as a solo developer, independently executing the full product lifecycle under competition time constraints.
- **OpenAI × Outskill AI Builders Hackathon**: Shortlisted in Top 1,000 developers nationwide to design, optimize, and ship a production-grade live AI product.
- **JEE Main 2024 – 98.16 percentile**: Top 1% nationally out of 1.2M+ candidates; qualified for JEE Advanced.
- **SRMC Mathematics Competition – AIR 27 (Finals)**: National finalist in the tournament jointly organized by IIT Bombay & IIT Madras.